



Warehouse Management System

Smart Inventory & Batch Sorting
Solution

| Problem Statement

The Challenge

Manual warehouse tracking is prone to severe data entry errors, leading to "Index Out of Bounds" logic failures during batch updates.

Current systems lack modularity, resulting in window clutter with excessive JFrame instances that cause memory overhead and user confusion.

Inconsistent data parsing between category names and IDs leads to critical database synchronization issues.

List of Users



Manager

Full system access, including Item Pricing, and master inventory overrides.



Warehouse Manager

Oversees arrived deliveries, generates sorting reports, and categorization.

Initial List of Tables

Table Name	Primary Key	Foreign Key(s)	Core Purpose
employee_accounts	emp_id	-	User authentication and credentials
category_table	cat_id	-	Item categorization (e.g., Tools, Const.)
delivery_info	batch_id	cat_id	Initial recording of arrived crates
sorted_batch	sorted_id	batch_id, cat_id	Refined batch data after sorting logic

| Functional Requirements

- ✓ **Authentication:** Secure login with employee credential verification.
 - ✓ **Delivery Processing:** Input and modification of batch delivery data.
 - ✓ **Sorting Logic:** Dynamic category assignment and batch ID reconciliation.
 - ✓ **Inventory Sync:** Real-time updates across multiple data modules.
 - ✓ **Modular Navigation:** CardLayout transition between Pricing and Inventory.
 - ✓ **Error Handling:** Robust validation for null inputs and array bounds.
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| Input List

User Inputs

- Employee Credentials (ID/Pass)
- Batch & Crate Identifiers
- Item Quantities (Integer)
- Category Dropdown Selection
- Search Query Keywords

System Triggers

- JTable Selection Indices
 - Action Performed Listeners
 - Mouse Hover Interactions
 - Database Query Result Sets
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| Output List

Visual Outputs

- Dynamic Inventory Data Tables
- Batch Arrival Summaries
- Navigation "Card" Switches
- Status Notification Banners

System Notifications

- Success/Confirmation Dialogs
 - Technical Error Message Popups
 - Console Debugging Logs
 - Updated SQL Records
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| Use-Case Summary

